Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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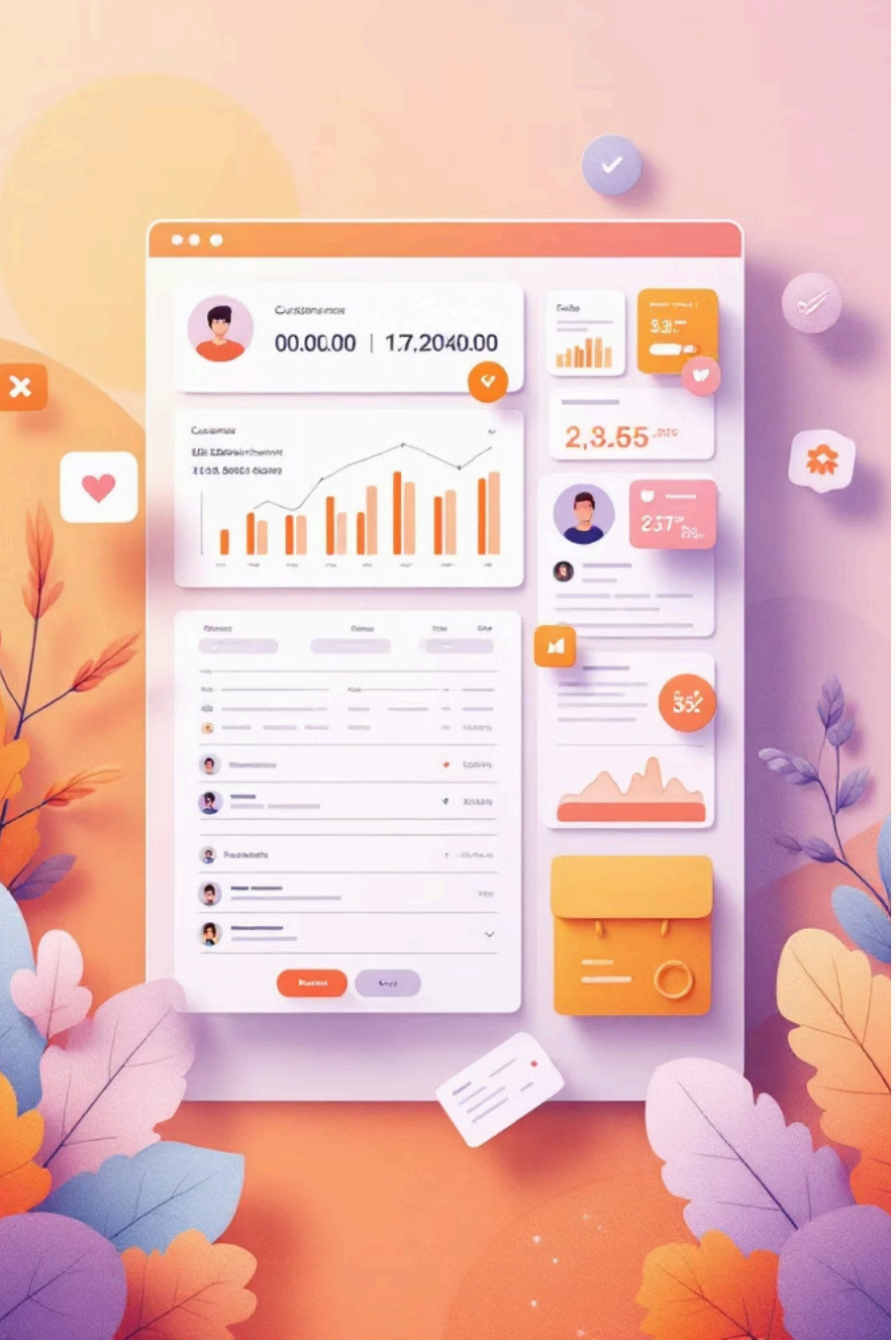

Dataset Overview

3.9K

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

Review ratings imputed

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Data Preparation Process

01

Data Loading

Imported using pandas

02

Exploration

Checked structure and statistics

03

Cleaning

Handled missing values

04

Standardization

Renamed to snake case

05

Engineering

Created age groups and frequency

06

Integration

Loaded to PostgreSQL

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Revenue by Gender

Key Finding

Male customers generate 2.1× more revenue than female customers

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Top Products by Rating

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

3.80 average rating

5

Skirt

3.78 average rating

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Shipping & Subscription Impact

Express Shipping

$60.48 average purchase vs. $58.46 standard

Non-Subscribers

73% of customers, $170K total revenue

Subscribers

27% of customers, $62.6K total revenue

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Discount Strategy Insights

Hats

Most discounted product

Coats

High discount rate

Sweaters

Seasonal promotions

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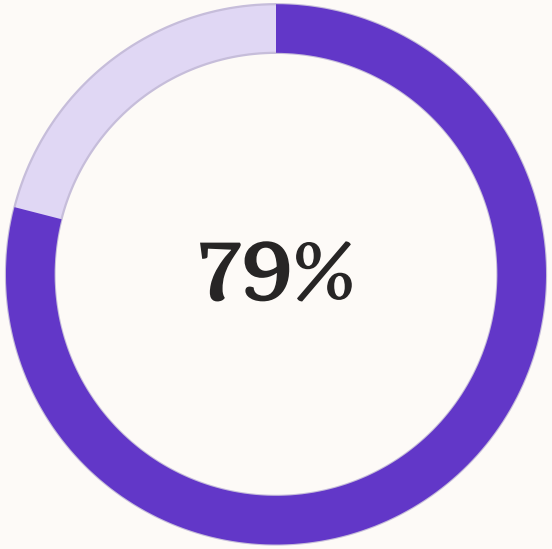
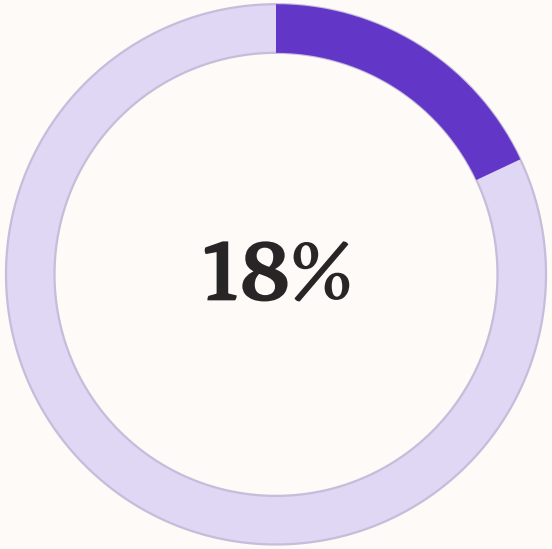
Customer Segmentation



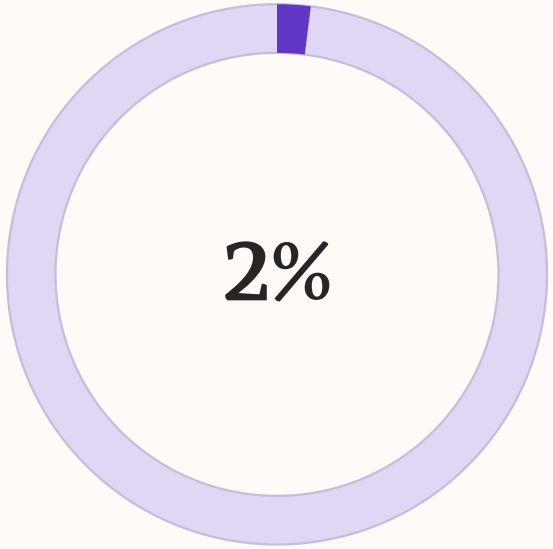
Loyal

3,116 customers



Returning

701 customers



New

83 customers

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Revenue by Age Group

Young adults drive highest revenue at $62,143

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move into loyal segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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